

Assignment 3 – Simple Shell with Pipes

Description:

This assignment requires us to create a simple shell. First, we need to handle command inputs without the piping. The input limit is 91 bytes. For each command we need to create a new process and execute it. Also, the simple shell should be able to terminate when exit is entered or EOF is reached. For the case with pipes, the program should be able to handle multiple pipes.

Approach:

First, I decided to write the program without piping. I will read the input and save it in the limited size buffer. Then I will keep each argument in the vector. For the execution of the command I need to create a new process, where the child process will execute and the parent process will wait for the termination of the child, to avoid orphans and zombies.

For piping, I need to begin with the same: copying the input into a fixed size buffer. Then I would differentiate the commands by | and keep them in the 2d vector. The first dimension is responsible for the pipes, and the second is responsible for the arguments. I would need file descriptors for each pipe for reading and writing. I need to create all the pipes beforehand. After that I would create a new process for each pipe, read the command from the first end of the pipe and write it to the second end and execute. Here also I would need to wait in the parent process.

Issues and Resolutions:

The first part where I just did the simple shell without piping went without major issues. One small bug I faced was that I didn't initialize the prompt from runoptions, so it did not display. I fixed it by taking the prompt from the argv array and printing it before each iteration. It worked well for manual entry, but behaves weirdly when reading from the file.

The major issues arose when I started working with pipes. It was challenging for me, because I had to learn the new functions in order to work with pipes. One of the issues I encountered is that I didn't close fds in the parent process. It resulted in the command not executing. It said the child process was executed, but no output was visible. I fixed it by closing all the read and write ends in the parent process.

```
student@student:~/CSC415/Assignment3/csc415-assignment-3-simpleshell-vincivv$ make run
gcc -c -o Almat_Dias_HW3_main.o Almat_Dias_HW3_main.c -g -I.
gcc -o Almat_Dias_HW3_main Almat_Dias_HW3_main.o -g -I. -l pthread
./Almat_Dias_HW3_main "Prompt> "
Prompt> cat commands.txt | wc
Child 11470 exited with status 0
```

Analysis:

Screen shot of compilation:

```
student@student:~/CSC415/Assignment3/csc415-assignment-3-simpleshell-vincivv$ make
gcc -c -o Almat_Dias_HW3_main.o Almat_Dias_HW3_main.c -g -I.
gcc -o Almat_Dias_HW3_main Almat_Dias_HW3_main.o -g -I. -l pthread
student@student:~/CSC415/Assignment3/csc415-assignment-3-simpleshell-vincivv$
```

Screen shot(s) of the execution of the program:

```
student@student:~/CSC415/Assignment3/csc415-assignment-3-simpleshell-vincivv$ ./Almat_Dias_HW3_main
prompt$
prompt$ ls -l
total 64
-rwxrwxr-x 1 student student 75744 Sep 25 23:05 Almat_Dias_HW3_main
-rw-rw-r-- 1 student student 5387 Sep 25 23:04 Almat_Dias_HW3_main.c
-rw-rw-r-- 1 student student 14416 Sep 25 23:05 Almat_Dias_HW3_main.o
-rw-rw-r-- 1 student student 66 Sep 25 21:51 commands.txt
-rw-rw-r-- 1 student student 1863 Sep 25 12:47 Makefile
-rw-rw-r-- 1 student student 7313 Sep 23 22:55 README.md
Child 11603 exited with status 0
prompt$ ls foo
ls: cannot access 'foo': No such file or directory
Child 11605 exited with status 2
prompt$ exit
student@student:~/CSC415/Assignment3/csc415-assignment-3-simpleshell-vincivv$
```

```
student@student:~/CSC415/Assignment3/csc415-assignment-3-simpleshell-vincivv$ make run < commands.txt
./Almat_Dias_HW3_main "Prompt> "
Almat_Dias_HW3_main Almat_Dias_HW3_main.c Almat_Dias_HW3_main.o commands.txt Makefile README.md
Prompt> Child 11622 exited with status 0
"Hello World"
Prompt> Child 11623 exited with status 0
total 80
drwxrwxr-x 4 student student 4096 Sep 25 23:05 .
drwxrwxr-x 3 student student 4096 Sep 23 22:55 ..
-rwxrwxr-x 1 student student 75744 Sep 25 23:05 Almat_Dias_HW3_main
-rw-rw-r-- 1 student student 5387 Sep 25 23:04 Almat_Dias_HW3_main.c
-rw-rw-r-- 1 student student 14416 Sep 25 23:05 Almat_Dias_HW3_main.o
-rw-rw-r-- 1 student student 66 Sep 25 21:51 commands.txt
drwxrwxr-x 7 student student 4096 Sep 23 22:56 .git
-rw-rw-r-- 1 student student 1863 Sep 25 12:47 Makefile
-rw-rw-r-- 1 student student 7313 Sep 23 22:55 README.md
drwxrwxr-x 2 student student 4096 Sep 24 18:29 .vscode
Prompt> Child 11624 exited with status 0
  PID TTY          TIME CMD
 11377 pts/1    00:00:00 bash
 11619 pts/1    00:00:00 make
 11620 pts/1    00:00:00 sh
 11621 pts/1    00:00:00 Almat_Dias_HW3_
 11625 pts/1    00:00:00 ps
Prompt> Child 11625 exited with status 0
Prompt> Child 11626 exited with status 0
 64      304
Child 11627 exited with status 0
ls: cannot access 'foo': No such file or directory
Prompt> Child 11628 exited with status 2
Prompt> Empty string entered: Nothing to do.
Prompt> student@student:~/CSC415/Assignment3/csc415-assignment-3-simpleshell-vincivv$
```